Windows PowerShell

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PS E:\IIIT G\SEM_5\CS351Lab\LAB9> python assignment_9.py
================================================================================
CS351 - IT Workshop II (Cloud Computing) | Lab Assignment 9
Student Name: Sayan Shit | Roll No: 2401179
================================================================================
[*] Dataset Path: E:\IIIT G\SEM_5\CS351Lab\LAB9\adult\adult.data
[*] Apache Spark session initialized successfully.
[*] Dataset loaded successfully. Total records: 32561

[*] Dataset Schema:
root
 |-- age: integer (nullable = true)
 |-- workclass: string (nullable = true)
 |-- fnlwgt: integer (nullable = true)
 |-- education: string (nullable = true)
 |-- education_num: integer (nullable = true)
 |-- marital_status: string (nullable = true)
 |-- occupation: string (nullable = true)
 |-- relationship: string (nullable = true)
 |-- race: string (nullable = true)
 |-- sex: string (nullable = true)
 |-- capital_gain: integer (nullable = true)
 |-- capital_loss: integer (nullable = true)
 |-- hours_per_week: integer (nullable = true)
 |-- native_country: string (nullable = true)
 |-- income: string (nullable = true)

[*] Preview of first 5 rows:
+---+----------------+------+---------+-------------+------------------+----------------+------------+-----+------+------------+------------+-------------
-+--------------+------+
|age|       workclass|fnlwgt|education|education_num|    marital_status|      occupation| relationship| race|   sex|capital_gain|capital_loss|hours_per_wee
k|native_country|income|
+---+----------------+------+---------+-------------+------------------+----------------+------------+-----+------+------------+------------+-------------
-+--------------+------+
| 39|       State-gov| 77516|Bachelors|           13|     Never-married|    Adm-clerical|Not-in-family|White|  Male|        2174|           0|           4
0| United-States| <=50K|
| 50|Self-emp-not-inc| 83311|Bachelors|           13|Married-civ-spouse|  Exec-managerial|      Husband|White|  Male|           0|           0|           1
3| United-States| <=50K|
| 38|         Private|215646|  HS-grad|            9|          Divorced|Handlers-cleaners|Not-in-family|White|  Male|           0|           0|           4
0| United-States| <=50K|
| 53|         Private|234721|     11th|            7|Married-civ-spouse|Handlers-cleaners|      Husband|Black|  Male|           0|           0|           4
```

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[*] Preview of first 5 rows:
+---+----------------+------+---------+-------------+------------------+-----------------+-------------+-----+------+------------+------------+-------------
-+--------------+------+
|age|       workclass|fnlwgt|education|education_num|    marital_status|       occupation| relationship| race|   sex|capital_gain|capital_loss|hours_per_wee
k|native_country|income|
+---+----------------+------+---------+-------------+------------------+-----------------+-------------+-----+------+------------+------------+-------------
-+--------------+------+
| 39|       State-gov| 77516|Bachelors|           13|     Never-married|     Adm-clerical|Not-in-family|White|  Male|        2174|           0|           4
0| United-States| <=50K|
| 50|Self-emp-not-inc| 83311|Bachelors|           13|Married-civ-spouse|  Exec-managerial|      Husband|White|  Male|           0|           0|           1
3| United-States| <=50K|
| 38|         Private|215646|  HS-grad|            9|          Divorced|Handlers-cleaners|Not-in-family|White|  Male|           0|           0|           4
0| United-States| <=50K|
| 53|         Private|234721|     11th|            7|Married-civ-spouse|Handlers-cleaners|      Husband|Black|  Male|           0|           0|           4
0| United-States| <=50K|
| 28|         Private|338409|Bachelors|           13|Married-civ-spouse|   Prof-specialty|         Wife|Black|Female|           0|           0|           4
0|          Cuba| <=50K|
+---+----------------+------+---------+-------------+------------------+-----------------+-------------+-----+------+------------+------------+-------------
-+--------------+------+
only showing top 5 rows

================================================================================
TASK 1: CLUSTERING ON AT LEAST 3 ATTRIBUTES (age, workclass, education, hours_per_week)
================================================================================
[*] K-Means Model Trained Successfully with k = 3
[*] Silhouette with squared Euclidean distance: 0.5201

[*] Cluster Centers (Standardized Space):
    Cluster 0: [ 0.20468775 -0.25340255  1.96493115 -0.07019425]
    Cluster 1: [-0.11843582 -0.41805851 -0.40190417 -0.0179798 ]
    Cluster 2: [ 0.30961116  2.00744121 -0.13983307  0.1419041 ]

[*] Cluster Size Distribution:
+----------+-----+
|prediction|count|
+----------+-----+
|         0| 4666|
|         1|21080|
|         2| 4979|
+----------+-----+
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[*] Cluster Size Distribution:
+----------+-----+
|prediction|count|
+----------+-----+
|         0| 4666|
|         1|21080|
|         2| 4979|
+----------+-----+

[*] Sample Clustered Records (Showing age, workclass, education, hours_per_week, prediction):
+---+----------------+---------+--------------+----------+
|age|workclass       |education|hours_per_week|prediction|
+---+----------------+---------+--------------+----------+
|39 |State-gov       |Bachelors|40            |2         |
|50 |Self-emp-not-inc|Bachelors|13            |1         |
|38 |Private         |HS-grad  |40            |1         |
|53 |Private         |11th     |40            |0         |
|28 |Private         |Bachelors|40            |1         |
|37 |Private         |Masters  |40            |1         |
|49 |Private         |9th      |16            |0         |
|52 |Self-emp-not-inc|HS-grad  |45            |1         |
|31 |Private         |Masters  |50            |1         |
|42 |Private         |Bachelors|40            |1         |
+---+----------------+---------+--------------+----------+
only showing top 10 rows

================================================================================
TASK 2(i): COUNTRY WITH HIGHEST NUMBER OF ADULTS (EXCEPT USA)
================================================================================
[*] Top 5 Non-USA Countries by Adult Count:
+--------------+-----------+
|native_country|adult_count|
+--------------+-----------+
|Mexico        |643        |
|Philippines   |198        |
|Germany       |137        |
|Canada        |121        |
|Puerto-Rico   |114        |
+--------------+-----------+
only showing top 5 rows
RESULT 2(i): The country with the highest number of adults (excluding USA) is 'Mexico' with 643 adults.
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```
only showing top 5 rows
RESULT 2(i): The country with the highest number of adults (excluding USA) is 'Mexico' with 643 adults.

================================================================================
TASK 2(ii): NUMBER OF PEOPLE WITH AT LEAST A MASTER'S WORKING IN 'Tech-support'
================================================================================
[*] Total count with at least Master's (education_num >= 14): 47

[*] Breakdown by specific education level in Tech-support:
+-----------+-------------+-----+
|education  |education_num|count|
+-----------+-------------+-----+
|Doctorate  |16           |3    |
|Prof-school|15           |7    |
|Masters    |14           |37   |
+-----------+-------------+-----+

[*] Additional Context:
    - Exactly Masters degree: 37
    - Masters + Prof-school + Doctorate (education_num >= 14): 47
RESULT 2(ii): Number of people with at least a Master's in Tech-support = 47

================================================================================
TASK 2(iii): NUMBER OF UNMARRIED MALES WHO WORK IN 'Local_govt'
================================================================================
[*] Detailed Breakdown of Marital Statuses for Males in Local Government:
+--------------------+--------------+-----+
|marital_status      |relationship  |count|
+--------------------+--------------+-----+
|Married-civ-spouse  |Husband       |859  |
|Never-married       |Not-in-family |126  |
|Never-married       |Own-child     |93   |
|Divorced            |Not-in-family |58   |
|Divorced            |Unmarried     |27   |
|Separated           |Not-in-family |16   |
|Never-married       |Unmarried     |16   |
|Never-married       |Other-relative|15   |
|Divorced            |Own-child     |9    |
|Widowed             |Not-in-family |7    |
|Widowed             |Unmarried     |6    |
|Separated           |Unmarried     |6    |
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|marital_status       |relationship  |count|
+---------------------+--------------+-----+
|Married-civ-spouse   |Husband       |859  |
|Never-married        |Not-in-family |126  |
|Never-married        |Own-child     |93   |
|Divorced             |Not-in-family |58   |
|Divorced             |Unmarried     |27   |
|Separated            |Not-in-family |16   |
|Never-married        |Unmarried     |16   |
|Never-married        |Other-relative|15   |
|Divorced             |Own-child     |9    |
|Widowed              |Not-in-family |7    |
|Widowed              |Unmarried     |6    |
|Separated            |Unmarried     |6    |
|Married-civ-spouse   |Other-relative|6    |
|Separated            |Own-child     |4    |
|Married-spouse-absent|Not-in-family |3    |
|Married-spouse-absent|Unmarried     |2    |
|Married-spouse-absent|Other-relative|2    |
|Divorced             |Other-relative|1    |
|Married-spouse-absent|Own-child     |1    |
|Married-civ-spouse   |Own-child     |1    |
+---------------------+--------------+-----+

[*] Counts under standard interpretations:
    - Based on relationship == 'Unmarried': 57
    - Based on marital_status == 'Never-married': 250
    - Based on all non-married marital statuses: 384

RESULT 2(iii):
    - If defining by relationship == 'Unmarried': 57
    - If defining by marital_status == 'Never-married': 250

================================================================================
ALL ASSIGNMENT 9 TASKS COMPLETED SUCCESSFULLY!
================================================================================
[*] Spark session stopped.
SUCCESS: The process with PID 23260 (child process of PID 22828) has been terminated.
SUCCESS: The process with PID 22828 (child process of PID 15036) has been terminated.
SUCCESS: The process with PID 15036 (child process of PID 25436) has been terminated.
PS E:\IIIT G\SEM_5\CS351Lab\LAB9>
```